S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

RiskPulse

AI/NLP Risk Engine with Event-Driven Portfolio Stress Testing (Module B)

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Problem & Approach

From headline to portfolio consequence, in one run

THE PROBLEM

Wholesale banking risk teams monitor thousands of headlines daily. Sanctions, downgrades and cyber incidents appear first as **unstructured text**, triaged by manual reading or keyword alerts. That leaves three gaps:

- Alerts are **not ranked** by severity
- Events are **not classified** in a consistent taxonomy
- Events are **not connected** to the bank's own exposures

What does this event do to our book?

APPROACH IN FIVE STEPS

1. **Ingest.** Google News RSS + GDELT DOC 2.0 (NewsAPI, CSV also supported) into one schema.
2. **Understand.** FinBERT sentiment, event class (domain terms + MiniLM), entities, learned impact score.
3. **Gate.** Only events with impact ≥ 7 trigger a stress scenario.
4. **Transmit.** Explicit shocks for equity, rates, spreads and default probability through a mixed portfolio.
5. **Present.** Signals, scenarios and position-level losses via API, dashboard and CSV export.

KEY DESIGN DECISIONS

Frozen FinBERT + MiniLM

Strong financial-text representations, reproducible inference, no training cost

Impact = next-day move

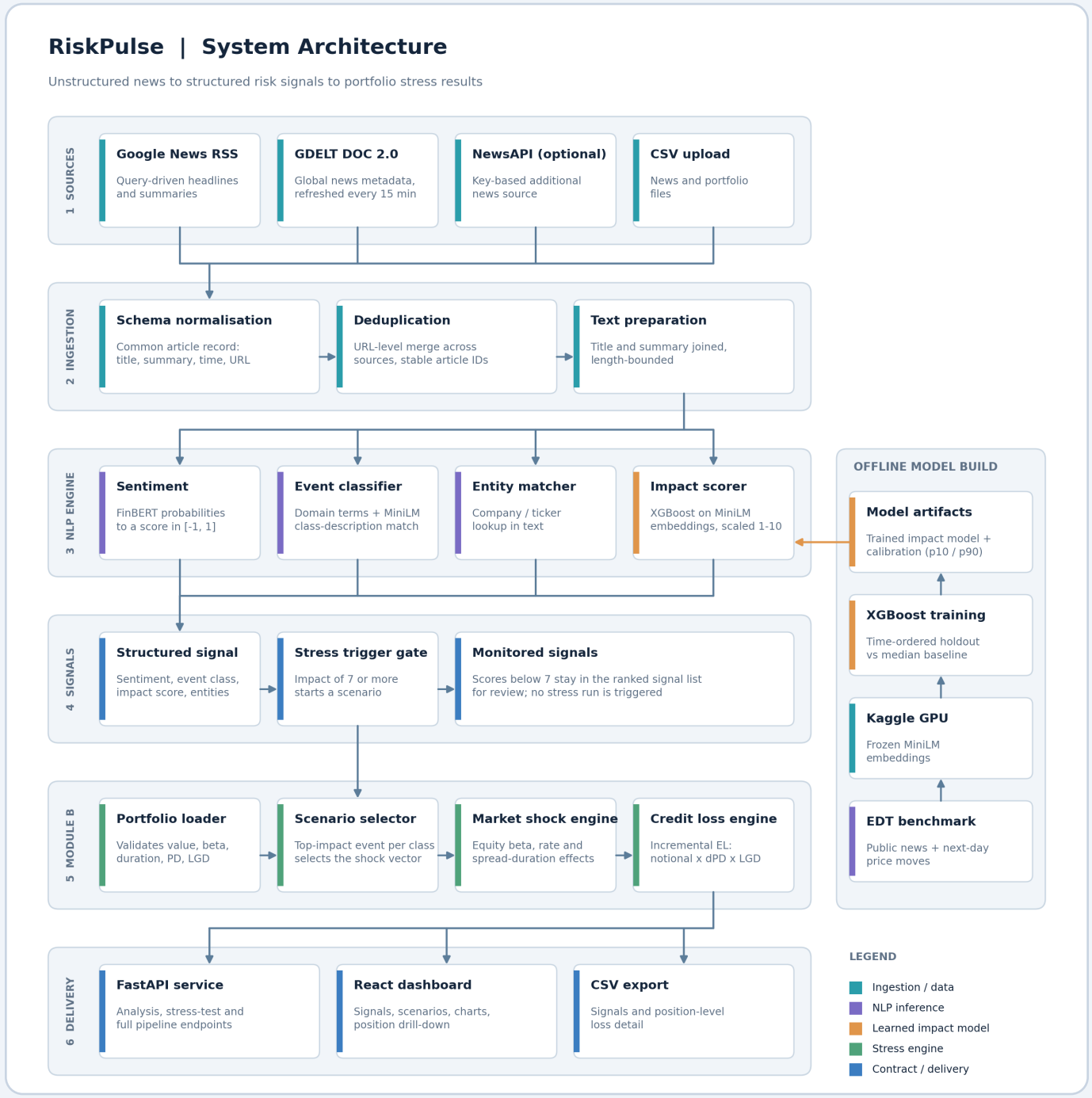
Predicted absolute price move scaled 1–10; severity tied to observed market reaction

Trigger: impact ≥ 7

Matches the case-study example; limits stress runs to material events

Explicit shock vectors

Transparent and auditable; every loss traces to a sensitivity



System Design

Data moves top to bottom through six layers

1 Sources
Google News RSS, GDELT DOC 2.0 (15-min refresh), NewsAPI (optional), CSV upload.

2 Ingestion
Common article record, URL-level dedup with stable IDs, length-bounded text.

3 NLP engine
Sentiment, event class, entities and impact score run in parallel on the same text.

4 Signals
One structured signal per article. Gate: impact ≥ 7 starts a scenario; below 7 stays monitored.

5 Module B
Portfolio loader, scenario selector, market shock engine, credit loss engine.

6 Delivery
FastAPI service, React dashboard with drill-down, CSV export.

Offline model build
EDT benchmark → Kaggle GPU (frozen MiniLM embeddings) → XGBoost training (time-ordered holdout vs median baseline) → model artifacts with p10 / p90 calibration, loaded by the impact scorer.

Implementation Highlights

Key modules, tech choices and why

Ingestion

One bounded request per source per run, merged by URL. Uploaded CSVs share the same schema (ID, time, source, publisher, title, summary, URL, query), so live and file inputs run identically.

Sentiment

Frozen **FinBERT** probabilities become a signed score in [−1, 1] plus a positive / neutral / negative label. A financial-lexicon scorer returns the same fields if the transformer isn't loaded, so the schema is stable everywhere.

Event classification

10 classes + Other: Geopolitical, Macro, Credit, M&A, Product Launch, Regulatory/Legal, Earnings, Market Liquidity, Cyber, Commodity/Supply Chain. Curated domain terms + **MiniLM** cosine similarity to class descriptions; every label has a confidence.

Impact score

XGBoost on MiniLM embeddings predicts the next-day absolute price move. Trained on the public EDT benchmark: 50,000 of 75,395 records, chronological 40,000 train / 10,000 validation, Kaggle GPU, log target. Scaled 1–10 between p10 (0.095%) and p90 (6.52%).

Module B stress engine

12 positions, **USD 105.12m**: loans, bonds, derivatives, equities, consumer credit. Per-class shock vectors (equity, rate, spread, PD×) scaled by k = s/10 when impact ≥ 7. Every loss traces to an explicit sensitivity.

Delivery & testing

FastAPI endpoints (analysis, stress test, full pipeline, portfolio, health); **React/Vite** dashboard with ranked signals, scenario selector, position drill-down; CSV export; optional LLM analyst. Tests cover the 7 trigger, below-trigger, input bounds and every asset type.

Loss calculation for each position i

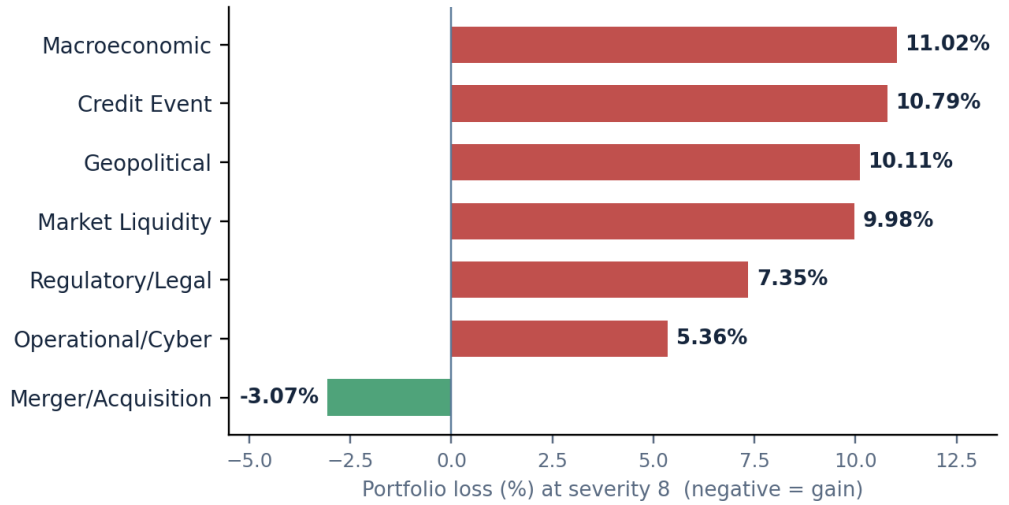
Market-shock factor	$f_i = 1 + \beta_i \Delta_{\text{eq}} - D_i \Delta_{\text{rate}} - S_i \Delta_{\text{spread}}$	Equity beta × equity return, minus rate duration × rate move, minus spread duration × spread move.
Incremental credit loss	$\Delta\text{EL}_i = N_i \left(\min(\text{PD}_i m_k,\ 0.95) - \text{PD}_i\right) \text{LGD}_i$	Notional × rise in default probability (capped at 95%) × loss given default.
Stressed value	$V_i^{\text{stressed}} = V_i f_i - \Delta\text{EL}_i$	Position value after the market shock, less the incremental credit loss (applied to loans, bonds, credit exposures).
Severity scaling	$k = s/10, \qquad m_k = 1 + (m-1)\,k$	Shocks scale with impact score s; m is the class PD multiplier. Active only for s ≥ 7; below 7 values are unchanged.

Key Results

Portfolio outcomes at severity 8 on the USD 105.12m book

Event class	After (USD m)	Loss (USD m)	Loss (%)
Macroeconomic	93.54	11.58	11.02
Credit Event	93.78	11.34	10.79
Geopolitical	94.49	10.63	10.11
Market Liquidity	94.63	10.49	9.98
Regulatory/Legal	97.39	7.73	7.35
Operational/Cyber	99.49	5.63	5.36
Merger/Acquisition	108.34	−3.22	−3.07

Before value USD 105.12m in every case. Negative loss = gain.



Impact model, validated out of time (10,000 later-dated rows)

XGBoost on MiniLM: **MAE 1.699 pp**, RMSE 3.224 pp
Train-median baseline: MAE 1.731 pp
Improves on the baseline on later-dated data.

vs. a keyword-alert workflow

Keyword alerts flag headlines with no severity order or portfolio consequence. RiskPulse returns, in one run: a **ranked signal list**, a driver event per class, the **stressed value** and the **most affected positions**.

Credit events

Losses concentrate in loans and bonds via the doubled PD multiplier and wide spread shock: L003 loses USD 2.09m; B001–B003 lose 1.53m, 1.45m, 1.22m.

Interpretable by class

Geopolitical events weigh most on equities; macroeconomic events act mainly through rate duration.

Monotone in severity

Loss rises steadily above the trigger (Macroeconomic ≈9.6% at 7 to ≈13.8% at 10). Positive corporate events give a net gain.

Domain Impact

Why it matters to a wholesale banking risk team

Faster, consistent triage

Analysts review a classified, severity-ranked list rather than raw feeds.

Exposure-aware alerts

Each trigger shows which loans, bonds, derivatives and equities are hit, and by how much.

Governance

Shocks, threshold and sensitivities are explicit. A risk officer can challenge or recalibrate them.

Reusability

The JSON/CSV signal contract serves other consumers (e.g. a sentiment-driven rebalancer) and an institution's own portfolio, with no engine changes.

	Keyword alerts	RiskPulse
Severity	None, flat alert list	Impact score 1–10, ranked
Classification	Keyword hits	10-class taxonomy + confidence
Portfolio link	None	Before / after value, position-level loss
Auditability	Opaque rules	Explicit shocks, threshold, sensitivities
Time to answer	Manual follow-up	One pipeline run: API, dashboard, CSV

Net effect

From unstructured text to a quantified answer: a ranked, classified signal, the scenario it triggers, and the positions it hurts, all traceable to explicit assumptions.

Limitations & Next Steps

Assumptions made explicit, with the planned extension for each

Area	Assumption / basis	Next step
Portfolio data	No organiser file supplied; 12-position wholesale-style portfolio generated, no client data	Any institutional CSV with the required columns loads unchanged
Scenario shocks	Analyst-defined shock vectors per event class, scaled by severity	Scenario analysis, not forecasts: calibrate against historical crises
Impact score	Proxy for single-stock next-day moves from the EDT benchmark	Not a causal loss estimate: add more periods and full-text features
Entity mapping	Name-to-ticker lookup over a compact public universe	Add validated issuer and counterparty mappings
News content	Headlines and summaries from public feeds; coverage varies	Add full-text retrieval and further sources
Trigger threshold	Impact ≥ 7, as in the case-study example	Make threshold and scaling configurable per desk
Scope	Module B implemented; Module A not built	Reuse the sentiment signal for a tactical rebalancer

GitHub **https://github.com/SHUSMIT/iitkgp_shusmit_hackathon**